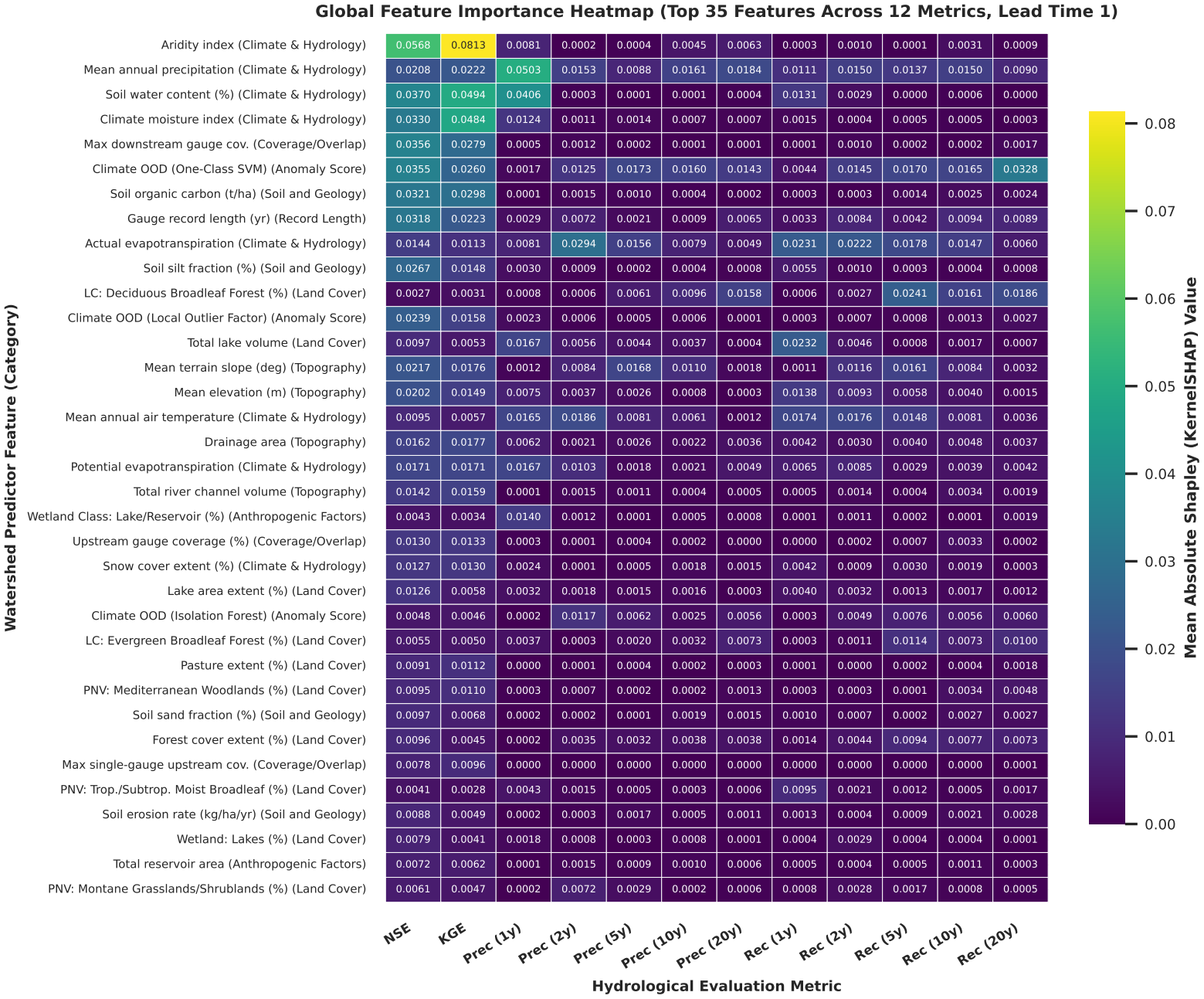
Global Feature Importance Heatmap (Top 35 Features Across 12 Metrics, Lead Time 1)
Watershed Predictor Feature (Category)
Aridity index (Climate & Hydrology) 0.0568 0.0813 0.0081 0.0002 0.0004 0.0045 0.0063 0.0003 0.0010 0.0001 0.0031 0.0009
Mean annual precipitation (Climate & Hydrology) 0.0208 0.0222 0.0503 0.0153 0.0088 0.0161 0.0184 0.0111 0.0150 0.0137 0.0150 0.0090
Soil water content (%) (Climate & Hydrology) 0.0370 0.0494 0.0406 0.0003 0.0001 0.0001 0.0004 0.0131 0.0029 0.0000 0.0006 0.0000
Climate moisture index (Climate & Hydrology) 0.0330 0.0484 0.0124 0.0011 0.0014 0.0007 0.0007 0.0015 0.0004 0.0005 0.0005 0.0003
Max downstream gauge cov. (Coverage/Overlap) 0.0356 0.0279 0.0005 0.0012 0.0002 0.0001 0.0001 0.0001 0.0010 0.0002 0.0002 0.0017
Climate OOD (One-Class SVM) (Anomaly Score) 0.0355 0.0260 0.0017 0.0125 0.0173 0.0160 0.0143 0.0044 0.0145 0.0170 0.0165 0.0328
Soil organic carbon (t/ha) (Soil and Geology) 0.0321 0.0298 0.0001 0.0015 0.0010 0.0004 0.0002 0.0003 0.0003 0.0014 0.0025 0.0024
Gauge record length (yr) (Record Length) 0.0318 0.0223 0.0029 0.0072 0.0021 0.0009 0.0065 0.0033 0.0084 0.0042 0.0094 0.0089
Actual evapotranspiration (Climate & Hydrology) 0.0144 0.0113 0.0081 0.0294 0.0156 0.0079 0.0049 0.0231 0.0222 0.0178 0.0147 0.0060
Soil silt fraction (%) (Soil and Geology) 0.0267 0.0148 0.0030 0.0009 0.0002 0.0004 0.0008 0.0055 0.0010 0.0003 0.0004 0.0008
LC: Deciduous Broadleaf Forest (%) (Land Cover) 0.0027 0.0031 0.0008 0.0006 0.0061 0.0096 0.0158 0.0006 0.0027 0.0241 0.0161 0.0186
Climate OOD (Local Outlier Factor) (Anomaly Score) 0.0239 0.0158 0.0023 0.0006 0.0005 0.0006 0.0001 0.0003 0.0007 0.0008 0.0013 0.0027
Total lake volume (Land Cover) 0.0097 0.0053 0.0167 0.0056 0.0044 0.0037 0.0004 0.0232 0.0046 0.0008 0.0017 0.0007
Mean terrain slope (deg) (Topography) 0.0217 0.0176 0.0012 0.0084 0.0168 0.0110 0.0018 0.0011 0.0116 0.0161 0.0084 0.0032
Mean elevation (m) (Topography) 0.0202 0.0149 0.0075 0.0037 0.0026 0.0008 0.0003 0.0138 0.0093 0.0058 0.0040 0.0015
Mean annual air temperature (Climate & Hydrology) 0.0095 0.0057 0.0165 0.0186 0.0081 0.0061 0.0012 0.0174 0.0176 0.0148 0.0081 0.0036
Drainage area (Topography) 0.0162 0.0177 0.0062 0.0021 0.0026 0.0022 0.0036 0.0042 0.0030 0.0040 0.0048 0.0037
Potential evapotranspiration (Climate & Hydrology) 0.0171 0.0171 0.0167 0.0103 0.0018 0.0021 0.0049 0.0065 0.0085 0.0029 0.0039 0.0042
Total river channel volume (Topography) 0.0142 0.0159 0.0001 0.0015 0.0011 0.0004 0.0005 0.0005 0.0014 0.0004 0.0034 0.0019
Wetland Class: Lake/Reservoir (%) (Anthropogenic Factors) 0.0043 0.0034 0.0140 0.0012 0.0001 0.0005 0.0008 0.0001 0.0011 0.0002 0.0001 0.0019
Upstream gauge coverage (%) (Coverage/Overlap) 0.0130 0.0133 0.0003 0.0001 0.0004 0.0002 0.0000 0.0000 0.0002 0.0007 0.0033 0.0002
Snow cover extent (%) (Climate & Hydrology) 0.0127 0.0130 0.0024 0.0001 0.0005 0.0018 0.0015 0.0042 0.0009 0.0030 0.0019 0.0003
Lake area extent (%) (Land Cover) 0.0126 0.0058 0.0032 0.0018 0.0015 0.0016 0.0003 0.0040 0.0032 0.0013 0.0017 0.0012
Climate OOD (Isolation Forest) (Anomaly Score) 0.0048 0.0046 0.0002 0.0117 0.0062 0.0025 0.0056 0.0003 0.0049 0.0076 0.0056 0.0060
LC: Evergreen Broadleaf Forest (%) (Land Cover) 0.0055 0.0050 0.0037 0.0003 0.0020 0.0032 0.0073 0.0003 0.0011 0.0114 0.0073 0.0100
Pasture extent (%) (Land Cover) 0.0091 0.0112 0.0000 0.0001 0.0004 0.0002 0.0003 0.0001 0.0000 0.0002 0.0004 0.0018
PNV: Mediterranean Woodlands (%) (Land Cover) 0.0095 0.0110 0.0003 0.0007 0.0002 0.0002 0.0013 0.0003 0.0003 0.0001 0.0034 0.0048
Soil sand fraction (%) (Soil and Geology) 0.0097 0.0068 0.0002 0.0002 0.0001 0.0019 0.0015 0.0010 0.0007 0.0002 0.0027 0.0027
Forest cover extent (%) (Land Cover) 0.0096 0.0045 0.0002 0.0035 0.0032 0.0038 0.0038 0.0014 0.0044 0.0094 0.0077 0.0073
Max single-gauge upstream cov. (Coverage/Overlap) 0.0078 0.0096 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0001
PNV: Trop./Subtrop. Moist Broadleaf (%) (Land Cover) 0.0041 0.0028 0.0043 0.0015 0.0005 0.0003 0.0006 0.0095 0.0021 0.0012 0.0005 0.0017
Soil erosion rate (kg/ha/yr) (Soil and Geology) 0.0088 0.0049 0.0002 0.0003 0.0017 0.0005 0.0011 0.0013 0.0004 0.0009 0.0021 0.0028
Wetland: Lakes (%) (Land Cover) 0.0079 0.0041 0.0018 0.0008 0.0003 0.0008 0.0001 0.0004 0.0029 0.0004 0.0004 0.0001
Total reservoir area (Anthropogenic Factors) 0.0072 0.0062 0.0001 0.0015 0.0009 0.0010 0.0006 0.0005 0.0004 0.0005 0.0011 0.0003
PNV: Montane Grasslands/Shrublands (%) (Land Cover) 0.0061 0.0047 0.0002 0.0072 0.0029 0.0002 0.0006 0.0008 0.0028 0.0017 0.0008 0.0005
NSE
KGE
Prec (1y)
Prec (2y)
Prec (5y)
Prec (10y)
Prec (20y)
Rec (1y)
Rec (2y)
Rec (5y)
Rec (10y)
Rec (20y)
Hydrological Evaluation Metric
Mean Absolute Shapley (KernelSHAP) Value
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00